

ODORHEIUL SECUIESC, Romania

WMO: 151680

Lat: 46.2967N

Lon: 25.2917E

Elev: 1720

StdP: 13.81

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	10.4	-0.2	5.8	9.7	4.1	7.2	13.7	18.5	32.4	16.3	30.9	2.1	20	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.5	86.6	67.1	83.9	66.0	81.3	65.3	69.3	82.3	68.0	80.3	66.6	78.5	4.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	99.5	73.6	63.8	94.5	72.7	62.5	90.3	70.9	34.5	82.4	33.3	80.1	32.2	79.0	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	13.4	10.5	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	27.5	32.9	39.2	48.8	58.3	65.3	66.7	67.5	58.5	50.2	41.5	31.2		
	DBStd	15.63	8.75	8.28	8.09	8.52	5.94	5.43	5.16	5.34	7.87	6.53	8.07	8.27		
	HDD50	2612	697	479	344	126	10	0	0	0	15	84	272	585		
	HDD65	6139	1162	898	799	487	217	62	44	37	220	459	705	1049		
	CDD50	2262	0	1	10	88	266	458	517	543	271	90	18	0		
	CDD65	316	0	0	0	0	9	70	96	115	26	0	0	0		
	CDH74	3298	0	0	1	35	163	712	881	1161	329	16	0	0		
	CDH80	861	0	0	0	0	15	160	239	370	77	0	0	0		
(14) Wind	WSAvg	3.6	2.8	3.2	4.0	4.8	4.0	3.7	3.7	3.5	3.8	3.3	3.4	2.9		
(15) Precipitation	PrecAvg	25.6	1.1	1.2	1.4	2.2	3.0	3.8	3.4	2.6	2.2	2.0	1.4	1.7		
	PrecMax	31.1	2.1	2.4	2.9	4.4	5.6	7.2	6.2	4.5	4.9	4.0	2.8	4.6		
	PrecMin	19.2	0.5	0.3	0.3	0.7	0.7	1.2	1.5	0.7	0.1	0.1	0.0	0.3		
	PrecStd	3.5	0.5	0.5	0.7	0.9	1.2	1.5	1.3	1.1	1.2	1.1	0.7	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	58.8	70.2	78.1	82.1	87.1	89.2	91.4	85.9	76.2	67.0	53.8		
		MCWB	43.6	48.8	52.2	57.3	60.4	69.7	67.9	68.1	64.6	58.4	53.4	46.3		
	2%	DB	44.9	53.2	63.2	73.9	78.7	84.2	85.8	87.4	81.6	72.6	62.3	47.9		
		MCWB	40.1	44.0	48.5	55.6	60.2	67.1	67.5	66.4	62.5	56.9	51.9	43.3		
	5%	DB	40.9	48.8	57.5	69.6	75.1	81.3	82.6	84.0	77.4	68.0	57.8	43.7		
		MCWB	37.2	41.9	46.4	53.9	59.9	65.9	66.4	65.9	60.9	55.3	49.3	40.2		
	10%	DB	37.8	44.2	51.9	64.5	71.2	78.0	79.2	80.8	73.1	62.8	53.5	40.4		
		MCWB	35.1	39.4	43.2	52.1	57.4	64.5	65.4	65.9	59.9	53.2	47.3	38.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	49.1	53.1	58.9	64.9	71.7	70.6	70.8	67.4	60.0	55.4	47.5		
		MCDB	48.9	57.4	68.8	74.1	75.3	85.4	83.0	84.7	82.8	72.6	64.5	51.9		
	2%	WB	40.2	45.2	49.8	57.0	62.4	68.9	69.0	69.2	64.4	58.2	53.0	43.7		
		MCDB	44.0	52.1	61.4	71.3	74.0	81.5	81.0	82.9	76.3	69.7	60.5	47.7		
	5%	WB	37.6	42.7	47.0	55.0	60.6	67.0	67.7	67.7	62.4	56.3	50.8	40.5		
		MCDB	40.5	47.7	56.3	67.9	71.9	78.7	79.2	80.4	74.3	65.9	57.2	43.1		
	10%	WB	35.4	39.8	44.4	52.9	58.9	65.4	66.4	66.3	60.8	54.0	48.0	38.3		
		MCDB	37.4	43.6	51.6	63.2	68.7	76.2	77.2	79.4	71.8	61.5	53.6	40.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	15.8	18.3	20.4	20.3	21.6	21.1	22.5	20.9	20.3	15.6	10.9		
		MCDBR	15.4	21.2	28.5	29.6	26.4	26.2	26.1	27.7	28.0	30.8	22.8	15.8		
	5% WB	MCWBR	10.7	13.1	14.6	13.7	12.0	11.5	10.7	10.9	11.3	16.1	13.2	11.4		
		MCWBR	13.6	19.7	24.8	26.6	23.4	24.6	23.2	25.1	24.8	27.8	20.1	14.4		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.331	0.373	0.428	0.408	0.405	0.425	0.432	0.391	0.354	0.330	0.304		
	taud		2.327	2.277	2.236	2.121	2.247	2.287	2.238	2.215	2.299	2.393	2.402	2.366		
	Ebn at Noon		249	264	268	262	273	273	266	258	259	253	237	235		
	Edh at Noon		24	31	38	46	42	41	43	42	35	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		417	686	1036	1379	1700	1841	1833	1680	1203	826	469	321		
	RadStd		52	77	118	139	157	157	154	148	131	101	57	39		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	+1.21	N/A	N/A	N/A	+1.49	N/A	N/A	-263	-362	+146	+34

Nomenclature: See separate page